CHARBEL TOUMIEH

Nationality: French ⋄ charbel.toumieh@epfl.ch

EDUCATION

Université Paris-Saclay Oct. 2019 - Sept. 2022
Ph.D. in robotics
MAV control, planning and SLAM in a high speed context - multi-agent planning.

Georgia Institute of Technology, Atlanta, USA Jan. 2018 - May 2019
MSc. Electrical and Computer Engineering.
Focus on robotics, control systems, computer vision and machine learning. *GPA: 4.0/4.0*

INP-ENSEEIHT, Toulouse, France Sept. 2015 - Jan. 2018
MSc. Electrical and Automation Engineering.
Transfer based on academic excellence. *Major de Promotion - Valedictorian*

Lebanese University, Faculty of Enginering II, Lebanon Sept. 2013 - June 2015
Engineering Common Core. *Top 5%*

EXPERIENCE

École Polytechnique Fédérale de Lausanne (EPFL) Feb. 2023 - present
Postdoctoral Fellow / Lecturer / Scientist

Working on aerial swarm autonomy as well as flapping-wing modeling and control. My responsibilities included:

- Research:
  - Developing new approaches for multi-agent quadrotor motion planning that beat the state-of-the-art in speed, robustness and smoothness.
  - Building a state-of-the-art swarm of drones (infrastructure) with 360 degrees vision and developing ROS2 software for control, motion planning, and perception.
  - Modelling and control of flapping-wing aerial vehicles, achieving state-of-the-art trajectory tracking performance.
  - Grants and proposal writing (secured a $280k Armasuisse Grant).
  - Supervising 4 PhD students and 15 Masters students in control, motion planning, reinforcement learning, and computer vision.
- Teaching:
  - Developing and lecturing the Aerial Robotics course (50% of the total lectures: attitude representations, cascaded control for multicopters and fixed wings, Kalman filtering, VIO/SLAM, motion planning and autonomous drone racing). Developing and supervising the exercises and hardware/software projects for autonomous drone racing.
  - Winner of the Teaching Assistant Award at EPFL awarded to the top 2% of assistants with 100% student satisfaction rate (92 students).

Parrot Oct. 2022 - Jan. 2023
Autonomous Flight Engineer

Implementation of state-of-the-art planning algorithms for industrial application of drones.

Université Paris-Saclay Oct. 2019 - Sept. 2022
Research Assistant

Design and implementation of state-of-the-art motion planning algorithms for single and multi-agent quadrotor systems. Won the tier 1 autonomous drone racing competition organized by Microsoft at NeurIPS 2019 using a newly developed motion planning algorithm that beats state-of-the-art methods.

Georgia Institute of Technology, Metz, France May 2018 - Nov. 2018
Research Assistant

Design of a VR application (using Unity3D) for 3D pointcloud visualization, preprocessing and surface reconstruction. Extracting 3D pointclouds from GPS and scale drifting measurements using optimization libraries (G2O).

ISAE-Supaero, Toulouse, France May 2017 - Sept. 2017
Research Assistant

Modeling, control, guidance and navigation of a quadcopter. Controllers design: classical controllers and state of the art numerical optimal controllers. Participated with the robotics team in the IMAV drone autonomy challenges: 1st prizes in the simulation challenge and the formation flight challenge.

AWARDS

Teaching Assistant Award - Engineering Department - EPFL 2025
Awarded to the top 2% of assistants in the department as voted by students.

Game of Drones - NeurIPS 2019
Won the tier 1 challenge in the drone racing competition that focused on motion planning and control - organized by Microsoft in partnership with Stanford and UZH-ETH.

SKILLS

Technical	Control Systems, Motion Planning, ML, ROS2, Robotics, CUDA, C++, Python
Linguistic	French (Fluent), English (Fluent), German (Intermediate), Arabic (Native)

HIGHLIGHTED PROJECTS

SwarmNxt - A Setup and Management System for Drone Swarms (IEEE - SSRR)
Developed a tutorial for the hardware and software setup of a swarm of drones with atomic instructions. Used automation script for a fast and parallelized software setup and update on all drones (Ansible). Implemented state-of-the-art control, motion planning, and perception algorithms:
`https://github.com/lis-epfl/swarm-nxt`

High-Speed Motion Planning for Aerial Swarms in Unknown and Cluttered Environments (IEEE - TRO)
Developed a motion planning algorithm for obstacle and intra-agent collision avoidance for a swarm of drones. Beats the state-of-the-art in velocity (2x improvement) and safety guarantees. Published in IEEE - Transactions on Robotics:
`https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10599811`
`https://github.com/lis-epfl/multi_agent_pkgs`

RESEARCH

`https://scholar.google.com/citations?user=34d6O30AAAAJ&hl=en&oi=ao`
`https://github.com/ctoumieh`